Infinite causal seeds: atomic capacity and entropy attainment

Proof completion for the finite-Shannon-theorem project

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Verification status. This document supplies the mathematical arguments for the measure and entropy completion. The accompanying Lean sources were compiled in the lake build of 11 October 2026 (Lean 4.35.0-rc3, mathlib 7d6757bc); no publication priority is asserted. They extend repository commit `0244f50`. The finite extraction and diagonal results already established there are stated explicitly below as inputs.

1 Definitions and the inherited inputs

Let T be a locally finite controlled history tree, with a finite nonempty set of actions at each history and a finite nonempty output set for each action. Kernels have positive probabilities; zero-probability outputs can be removed. Terminal histories are padded deterministically. Let $p(h)$ be the product of the transition probabilities along a valid history h, and zero on invalid raw histories. Thus $p(\varnothing) = 1$ and $\sum_y p(hay) = p(h)$ for every available action a.

A deterministic strategy gives an output at every history–action pair. Its space $\mathcal{S}$ is a countable product of finite discrete spaces, hence compact and metrizable. Let E_h be the clopen set of strategies agreeing with h. An exact causal law is a Borel probability μ on $\mathcal{S}$ satisfying $\mu(E_h) = p(h)$ for every h.

For the transcript distribution $P_{n,\pi}$ of a deterministic policy at depth n, set

$$A_K(P) = \max_{|D|\le K} P(D), \qquad a_K = \inf_{n,\pi} A_K(P_{n,\pi}), \qquad q = \sup_K a_K \in [0,1].$$

An ordinary source has an exact transcript readout for each (n,π), without any compatibility requirement between the readouts.

The repository's finite extraction and diagonal construction give weights $w_i \ge 0$ and strategies $s_i \in \mathcal{S}$, $i \in \mathbb{N}$, such that

$$\sum_i w_i = q, \qquad c(h) := \sum_i w_i \mathbf{1}_{E_h}(s_i) \le p(h). \tag{1}$$

In its notation these are `diagMass_eq_tsum_atomQ` and `cylMass_le_path`. Also $r(h) := p(h) - c(h) \ge 0$, and

$$r(\varnothing) = 1 - q, \qquad \sum_y r(hay) = r(h). \tag{2}$$

These are the only inherited nontrivial inputs to the atomic completion. The entropy argument in Section 3 does not require this particular greedy sequence.

2 The missing measure and the sharp converse

Theorem 1 (Attained atomic capacity)**.** *The maximal atomic mass of an exact causal source and that of an ordinary source are both q. In particular, an exact countable causal seed exists if and only if $q = 1$.*

Proof. Define $\nu = \sum_i w_i \delta_{s_i}$. Countable additivity and (1) give $\nu(\mathcal{S}) = q$ and $\nu(E_h) = c(h)$; repeated strategies simply accumulate their weights at the same point.

For every history–action coordinate, use the probability distribution

$$k^r(y \mid h, a) = \begin{cases} r(hay)/r(h), & r(h) > 0, \\ k(y \mid h, a), & r(h) = 0, \end{cases}$$

where k is the original output kernel. Equation (2) proves normalization in the first case; the second is already normalized. Take the independent product λ of these coordinate laws on $\mathcal{S}$. A valid path constrains distinct coordinates, so $\lambda(E_h)$ is the product of its successive k^r factors. Induction on its length gives

$$(1 - q)\lambda(E_h) = r(h). \tag{3}$$

Indeed, if the parent residual is positive the next factor is the required ratio. If it is zero, every child residual is zero by nonnegativity and (2), and both sides remain zero. This also covers $q = 1$. Invalid histories have an empty cylinder and zero path mass. Therefore

$$\mu := \nu + (1 - q)\lambda \quad \text{satisfies} \quad \mu(\mathcal{S}) = 1, \qquad \mu(E_h) = c(h) + r(h) = p(h).$$

Since $\mu \geq \nu$, its atomic mass is at least q.

For the converse, let η be any ordinary source law with measurable singletons and exact transcript readouts. A finite set F of K source points maps, under each readout, into at most K transcripts. Hence

$$\eta(F) \leq A_K(P_{n,\pi}) \quad \text{for every } n, \pi, \qquad \eta(F) \leq a_K \leq q.$$

Taking the supremum over finite F gives $\mathrm{At}(\eta) := \sum_x \eta(\{x\}) \leq q$. This applies to μ as well, so $\mathrm{At}(\mu) = q$.

A source and its full tuple of transcript readouts yield the same maximum: the pushforward tuple law has atomic mass at least that of the source, and the preceding upper bound also applies to the tuple law. Thus the ordinary-coupling maximum is attained too. If $q = 1$, ν alone is an exact probability supported on the countable set $\{s_i : i \in \mathbb{N}\}$. Conversely, a countable exact probability has atomic mass one, so the upper bound forces $q = 1$. □

3 Entropy: no escape of mass

Use natural logarithms in this section. For a countable seed with weights v_i, set $H(v) = \sum_i -v_i \ln v_i \in [0, \infty]$, with $0 \ln 0 = 0$. Let C_n be the infimum of seed entropy among sources exact through depth n, and C_∞ the infimum among seeds exact at every finite depth. The infimum of an empty class is $+\infty$.

We may enumerate positive masses in decreasing order and pad a finite list with zeros. Here is why this is legitimate even for an infinite support. Every nonempty remainder of a summable sequence of positive masses has a largest element: its positive supremum is attained among the finitely many entries exceeding half that supremum. Successive selection of a largest entry exhausts the support, since only finitely many masses exceed any fixed positive threshold. Relabeling and deleting zero-mass labels preserve entropy and all readouts.

At a finite horizon these definitions agree with optimization over a finite distribution on finite strategies. Restriction to depth n maps any seed into the finite set of depth-n strategies. Merging equal strategies cannot increase entropy: for a group of masses of total p,

$$\sum_j -v_j \ln v_j = -p \ln p + p \sum_j -\frac{v_j}{p} \ln \frac{v_j}{p} \geq -p \ln p$$

when $p > 0$, and both sides are zero when $p = 0$. Conversely, any finite strategy extends by arbitrary outputs beyond depth n. Thus these are the original finite-horizon costs, not a stronger feasibility requirement. The Lean file uses the ranked, zero-padded model as its explicit definition; the preceding relabeling and finite-quotient identifications are supplied here as mathematical arguments, not as additional compiled declarations.

Lemma 2 (Uniform tail bound)**.** *If $v_0 \geq v_1 \geq \cdots \geq 0$, $\sum_i v_i = 1$, and $H(v) \leq U < \infty$, then for every integer $K \geq 1$,*

$$\sum_{i \geq K} v_i \leq \frac{U}{\ln(K+1)}.$$

Proof. $(i+1)v_i \leq \sum_{j=0}^{i} v_j \leq 1$. Hence, for $i \geq K$ with $v_i > 0$, $-\ln v_i \geq \ln(K+1)$. Multiply by v_i and sum. Zero terms cause no difficulty. All entropy terms are nonnegative. □

Lemma 3 (Exact compactness with an entropy bound)**.** *Suppose seed j has ranked weights $v_i^{(j)}$ and strategy labels $s_i^{(j)}$, is exact through depth j, and $H(v^{(j)}) \leq b_j$, where $b_j \to B < \infty$ and $b_j \leq U < \infty$. Then there is a seed exact at all depths with entropy at most B.*

Proof. The space $[0,1]^{\mathbb{N}} \times \mathcal{S}^{\mathbb{N}}$ is compact metrizable. Pass to a subsequence $j_m \to \infty$ on which, for each label i, $v_i^{(j_m)} \to v_i$ and $s_i^{(j_m)} \to s_i$. The limiting weights remain nonnegative and decreasing. Each finite head has mass at most one, so $\sum_i v_i \leq 1$. Lemma 2 gives, for every fixed $K \geq 1$,

$$1 - \sum_{i<K} v_i = \lim_m \left(1 - \sum_{i<K} v_i^{(j_m)}\right) \leq \frac{U}{\ln(K+1)}.$$

Letting $K \to \infty$ proves $\sum_i v_i = 1$. This is the essential no-escape step; coordinatewise convergence alone would not suffice.

Fix a history h. Its cylinder is clopen, so for each fixed i the indicators $\mathbf{1}_{E_h}(s_i^{(j_m)})$ eventually equal $\mathbf{1}_{E_h}(s_i)$. For every K, exactness at depth $j_m \geq |h|$ gives

$$\sum_{i<K} v_i^{(j_m)} \mathbf{1}_{E_h}(s_i^{(j_m)}) \leq p(h),$$
$$\sum_{i<K} v_i^{(j_m)} (1 - \mathbf{1}_{E_h}(s_i^{(j_m)})) \leq 1 - p(h).$$

Pass first to $m \to \infty$, then to $K \to \infty$. The two limiting series sum to one; their upper bounds also sum to one. Both inequalities are therefore equalities. In particular, $\sum_i v_i \mathbf{1}_{E_h}(s_i) = p(h)$ for every h.

Finally, $x \mapsto -x \ln x$ is continuous on $[0,1]$. For each K,

$$\sum_{i<K} -v_i \ln v_i = \lim_m \sum_{i<K} -v_i^{(j_m)} \ln v_i^{(j_m)} \leq \lim_m b_{j_m} = B.$$

Taking the supremum over K proves $H(v) \leq B$. No continuity of entropy on the full countable simplex has been used. □

Theorem 4 (Projective entropy identity and attainment)**.** *For every locally finite controlled tree,*

$$\boxed{C_\infty = \sup_{n \geq 0} C_n.}$$

If this value is finite, one exact countable seed attains it.

Proof. Restriction gives $C_n \le C_{n+1} \le C_\infty$. This proves the result if $\sup_n C_n = +\infty$. Otherwise put $B = \sup_n C_n < \infty$. By the definition of infimum, choose a depth-n exact seed with

$$H(v^{(n)}) < C_n + \frac{1}{n+1} \le B + \frac{1}{n+1}.$$

There is no need to assume existence of finite-horizon minimizers. Extend each finite strategy if necessary and rank its weights. Apply Lemma 3 with $b_n = B + 1/(n+1)$ and $U = B + 1$. The resulting infinite seed has entropy at most B. Restriction supplies the opposite inequality, proving both equality and attainment. □

Corollary 5 (Connection to the finite Shannon bound)**.** *Return to bits and let L_n denote the finite adaptive-envelope entropy. If the finite theorem gives $L_n \le C_n \le L_n + c_*$ with $c_* = (1 + \log_2 e)/2$, then*

$$L_\infty \le C_\infty \le L_\infty + c_*, \qquad L_\infty := \sup_n L_n.$$

In particular, $L_\infty < \infty$ if and only if a finite-entropy exact seed exists on the whole tree. This condition implies $q = 1$.

Proof. Changing logarithm base multiplies all entropy costs by $1/\ln 2$. Take suprema in the finite bounds and use Theorem 4. Attainment provides the seed when the bound is finite. Conversely, any finite-entropy exact seed bounds every L_n by restriction. Its countable support and Theorem 1 imply $q = 1$. □

4 What is included in the formalization candidate

- `MeasureCompletion.lean`: residual product measure, exact mixture, cylinder probabilities and probability normalization.
- `AtomicCapacity.lean`: arbitrary-source upper bound, attainment, and the countable-seed equivalence $q = 1$.
- `EntropyLimit.lean`: ranked-seed definitions, tail estimate, compactness, projective entropy identity, attainment, and the exact probability measure of the optimal seed. Entropy is in nats in this file.

The explicit Lean costs optimize over ranked countable seeds. The proof of their identification with differently encoded finite optimization problems is given above in prose. The envelope corollary combines the projective identity with the repository's finite theorem; it is not an additional Lean declaration in this package. None of these new files has been compiled here. Successful compilation and inspection of `#print axioms` remain required before claiming a formal certificate.

References

[1] J. Mikulík, *finite-Shannon-theorem*, repository, commit `0244f50`. `https://github.com/jendamikulik/finite-Shannon-theorem`.